



Deep learning solutions for audio event detection in a swine barn using environmental audio and weak labels

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Accepted: 8 April 2025 / Published online: 17 April 2025

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Abstract

The increasing demand for animal protein products has led to the emergence of Precision Livestock Farming (PLF) and the adoption of sensing technologies, big data solutions, and Machine Learning (ML) methods in modern livestock farming. At the same time, the audio signal processing field has undergone notable advancements in recent years, transitioning from traditional techniques to more sophisticated ML approaches, with open challenges in detecting and classifying complex, low-quality, and overlapping sounds in real-world scenarios. In this paper, we evaluate deep learning methods, conceived from computer vision to attention-based approaches, for Audio Event Detection (AED) on a novel audio dataset from a swine farming environment with challenging characteristics, such as weak annotations and high amounts of noise. The primary purpose of our study is to prospect effective AED solutions for the development of tools for auditing livestock farms, which could be used to improve animal welfare. Our results show that, despite inherent limitations in the dataset's size, class imbalance, and sound quality, Convolutional Neural Network (CNN) and attention-based architectures are respectively effective and promising for detecting complex audio events. Further research may explore avenues for optimizing model performance in similar, real-life datasets while simultaneously amplifying annotated events and reducing annotation costs, thereby enhancing the broader applicability of AED methods in diverse audio processing scenarios.

Keywords AED · Audio classification · Machine learning · Computer vision · Precision livestock farming · Weak annotations

1 Introduction

Recent studies predict a significant increase in the global demand for animal-based products in the coming decades [1, 2], leading to higher production efficiency requirements. Consequently, the livestock industry has adopted Precision Livestock Farming (PLF) techniques to improve productivity and animal welfare. This approach involves using sensing technologies, big data systems, and ML methods. In livestock, ML techniques have been successfully applied in various fields [3–7].

The field of audio signal processing has undergone notable advancements in recent years, transitioning from traditional techniques, e.g., Gaussian Mixture Models (GMMs), Hidden Markov Models (HMMs), and Support Vector Machines (SVMs) [3, 8–11], to more sophisticated Machine Learning (ML) approaches. These developments parallel broader breakthroughs in deep learning, which have demonstrated

remarkable promise beyond audio domains, such as medical image analysis for Computed Tomography (CT) image segmentation and body composition assessment [12], as well as breast cancer histopathology image classification [13]. Recent work in the audio domain has delved into Automatic Speech Recognition (ASR) [14], Environmental Sound Classification (ESC) [15], and numerous other applications, becoming increasingly relevant in a variety of real-world scenarios.

In livestock scenarios, audio analysis can be of considerable importance, as sound monitoring in livestock farming environments could be applicable in assessing the regularity and quality of human handling, animal behaviors, and animal health, which may improve animal welfare. Surveillance systems can collect audio data unobtrusively and inexpensively, enabling the discovery of relevant knowledge correlating biological aspects and the sound itself. A notable trait of surveillance livestock sounds in Brazil, for instance, is that environments are generally open, thus generating large amounts of noise with multiple sound sources. Despite the

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array of potential advancements, until recently, few studies in livestock environments have explored deep learning methods with audio data [16–19]. In AED, there are still challenges in detecting and classifying complex, low-quality, and overlapping sounds in real-world scenarios, which require the development of more robust and accurate ML models that can adapt to the variability and complexity of real-world audio data [15].

In this paper, we aim to evaluate deep learning methods for AED on a novel audio dataset from a swine farming environment. To the extent of our knowledge, this is the first study that integrates animal, human, and environmental sounds for AED and auditing purposes. As such, the main contributions of our work are:

- An evaluation of computer vision and attention-based methods for the AED task with a novel dataset.
- Extensive experiments to evaluate the impact of different ML techniques on AED on the proposed dataset.

This paper is organized as follows: Section 2 briefly describes the background and recent related studies in audio mining, AED, and applications in livestock. Section 3 presents our novel audio dataset and its characteristics. Section 4 describes the methodology for experiments we conducted in this study. Finally, in Section 5, we present and discuss the obtained results, with an outline of the insights and contributions of this research work.

2 Background and related work

In the past, traditional methods for Audio Event Detection (AED) primarily relied on using features such as Mel-Frequency Cepstral Coefficients (MFCCs) and classifiers based on GMMs, HMMs, and SVMs [3, 8–11]. While these techniques are relatively simple and require less time for training, they may struggle to handle more complex tasks with larger datasets [20]. More recently, studies have turned to deep learning methods for AED, drawing inspiration from Computer Vision and Natural Language Processing (NLP). Techniques like Convolutional Neural Networks (CNNs) [21–25] and attention-based models [25, 26] have enabled state-of-the-art performance, but they require more computing power, longer training times, and larger datasets [20].

In general scenarios, a few studies investigated the usefulness of widespread deep learning architectures on audio tasks. One example is that of Hershey et al. [21], which examined the effectiveness of various deep architectures, such as AlexNet [27], VGG [28], Inception V3 [29], and ResNet [30], on audio classification with the AudioSet [31]

dataset. The authors found that vision models can effectively classify audio events in large datasets with few architectural adaptations.

To better capture long-range global contexts, recent research work added self-attention mechanisms on top of CNNs, creating CNN-attention hybrid models [32, 33]. To investigate whether CNNs are essential for audio classification, Gong et al. [26] proposed the Audio Spectrogram Transformer (AST) architecture, which is a pure attention-based model, i.e., without convolutional layers for feature extraction, inspired by the Vision Transformer (ViT) architecture [34]. That study evaluated the AST on audio classification benchmarks, achieving state-of-the-art results despite requiring pre-training before training on smaller datasets.

3 Dataset

With increasing interest in utilizing ML techniques in precision livestock, researchers from the School of Veterinary Medicine and Animal Science (FMVZ) of the University of São Paulo (USP) performed various studies with sensing technologies across multiple animal species and environments. The Swine dataset originates from a study regarding the provision of a feed additive to groups of finishing pigs, aiming to analyze performance, behavioral changes, and other aspects. To this end, FMVZ's researchers created an infrastructure for 24/7 monitoring and storage in video format (.mp4), which we convert to raw waveform (.wav) with a sample rate of 16KHz.

The original Swine dataset contains 12 143 6-minute files captured with a Foscam FI8918W IP Camera between September 3, 2020, and October 13, 2020, totaling 1 214.3 hours. Due to high labeling costs, the Annotated Swine (aSwine) dataset contains only approximately 15 hours of data. Our annotation methodology employs a *weak* labeling approach where, for each 5-second interval, one or more labels are assigned based on the presence/absence of associated events.

Annotation Process and Inter-Annotator Agreement

The labeling task was performed by a primary domain specialist with experience in swine production. Then, two additional animal production experts independently reviewed random subsets of the annotations to validate correctness and reduce subjectivity. Potential disagreements were discussed in joint sessions to reconcile label definitions (e.g., whether a short grunt qualified as “stress” or was merely background noise). While some classes (e.g., Others (H)) were more prone to subjective interpretation, these joint reviews helped ensure that subtle events, such as certain animal vocalizations and brief human interactions, were labeled accurately.

It is important to emphasize that the audio annotations were guided by both the sound and the synchronized video recordings of the same event, ensuring accuracy and consistency throughout the annotation process.

3.1 Dataset characteristics

The aSwine events fall into two major categories: 1) animal and 2) human events, as in Table 1. The category and class distributions are illustrated in Figs. 1 and 2, respectively. The labels we selected capture pertinent events for observing human handling and animal behaviors, in which detection could facilitate auditing in day-to-day operations. In the annotation process, we included a class labeled “Other Events” for each category (denoted as “Others (H)” and “Others (A)”) to encompass occurrences not covered by existing labels yet potentially significant in detection. Examples of such events may involve chirping birds, wild animal calls, unidentified loud noises, and automobile sounds. This category accounts for diverse and unforeseen events potentially relevant in the detection context.

Mitigation of Class Imbalance, Noise, and Overlapping Events

We emphasize that, in this initial study, we did not employ data augmentation, oversampling, or noise reduction methods. Instead, we designed and evaluated our deep learning models under the raw, noisy conditions of the aSwine dataset, aiming to assess how well different architectures could cope with these challenges out-of-the-box. Future work could explore specialized techniques to counteract class imbalance (e.g., class-weighted losses or focal loss) and data augmentation strategies (e.g., time-frequency masking) or noise reduction methods to enhance performance further. Although our experiments already adopt a multilabel training approach to accommodate overlapping events, further refinement of annotation schemes may help reduce ambiguity in cases involving highly concurrent or subtle sound sources.

We showcase examples of 1s audio segments and their respective spectrograms in Fig. 3, where we can observe different frequency patterns for *Human Speech* and *Stress/Disputes* events. Notably, stress sounds tend to exhibit higher energy around 2kHz than human speech. These

Table 1 Categories and labels in the aSwine dataset

Human	Animal
Cleaning	Stress/Disputes
Speech	Coughing/Sneezing
Feed Provision	Others (A)
Others (H)	

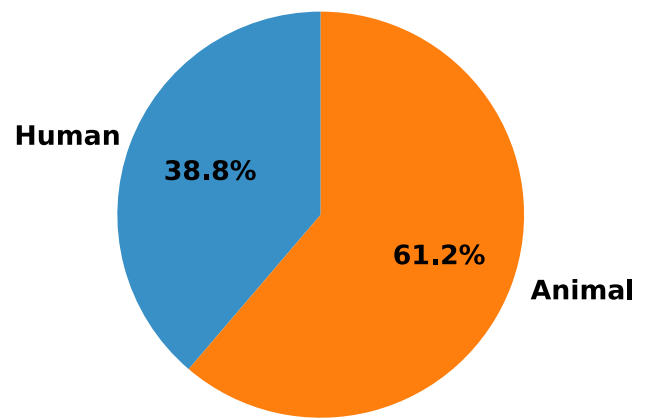


Fig. 1 Distribution of labels by category of the aSwine dataset

unique spectral profiles exemplify why domain-specific algorithms or targeted data augmentation strategies could be particularly beneficial here.

As indicated in Figs. 1 and 2, the aSwine dataset is class-imbalanced, with fewer instances of *Feed Provision* or *Others (H)*. Moreover, the open-environment recordings exhibit considerable noise. Multiple, possibly overlapping audio sources at varying distances from the sensor contribute to differences in sound intensities even within the same event class. Such challenges, familiar to many real-life agricultural/livestock datasets but less emphasized in general-purpose corpora, demand careful consideration of ML techniques for AED.

Overall, the aSwine dataset presents valuable opportunities to investigate real-world AED tasks in livestock production. The noisy conditions, a wide range of sound intensities, and overlapping events reflect the complexities practitioners face daily. By highlighting these domain-specific characteristics and inter-annotator validation, we aim to provide a solid foundation for robust event-detection models that can generalize to other noisy livestock environments. We intend that subsequent work may build upon these findings by introducing additional techniques that specifically target class imbalance, noise, and overlapping sounds in precision livestock audio data.

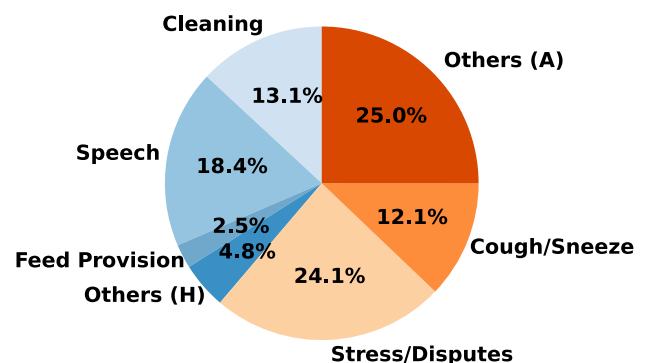


Fig. 2 Distribution of labels by class of the aSwine dataset

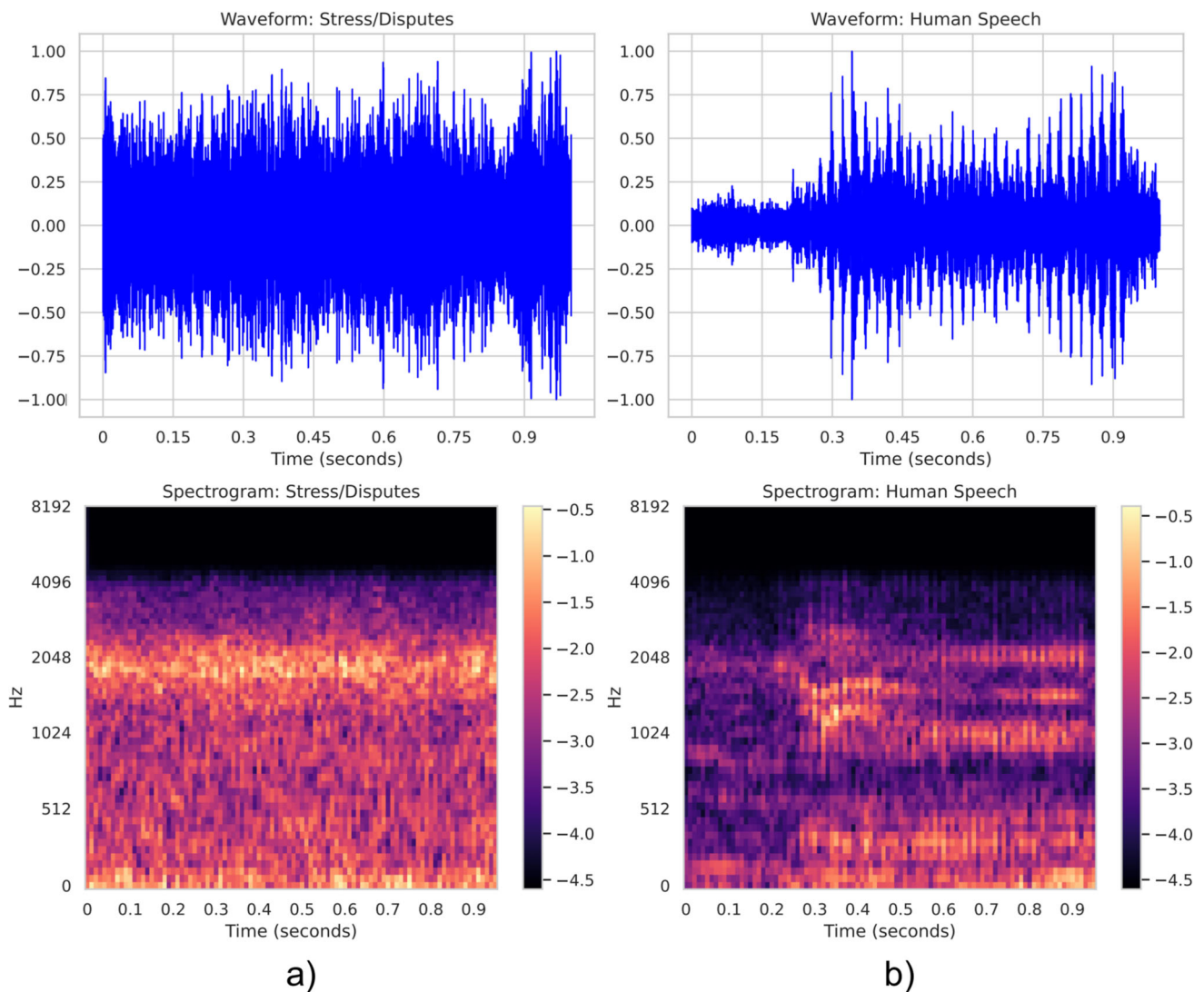


Fig. 3 Examples of waveforms and log-mel spectrograms from two different classes: a) Stress/Disputes; and b) Human Speech

3.2 Comparison with existing AED datasets

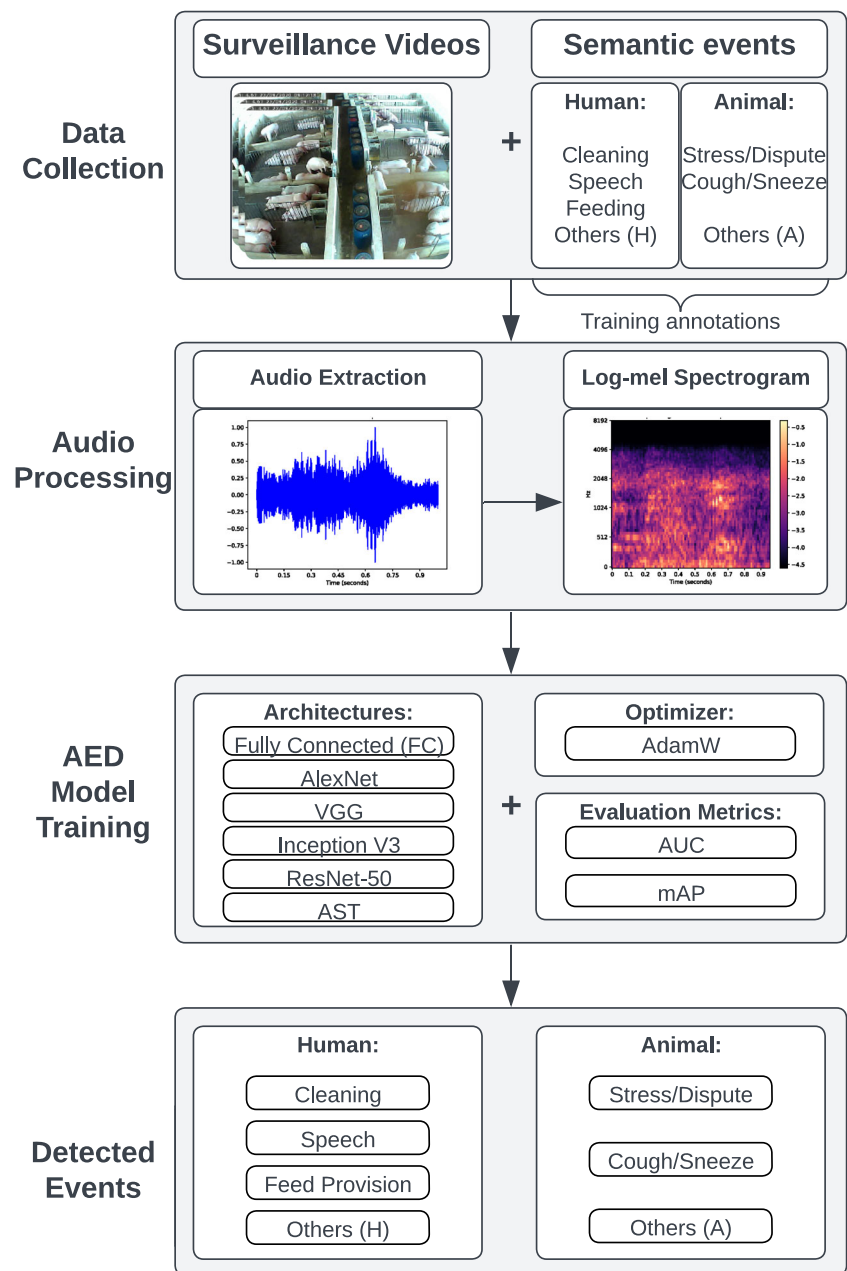
Compared to well-known general-purpose AED datasets such as UrbanSound8K [35] or AudioSet [31], aSwine offers several domain-specific challenges. First, our recordings are acquired in an *open swine barn environment* using low-cost microphones; consequently, the audio is often noisy with overlapping sources and fluctuating intensities. Second, the *weak labeling* approach, combined with a relatively small labeled subset (15 hours vs. hundreds or thousands of hours in AudioSet), introduces increased difficulty for training models to distinguish fine-grained events in short time windows. Third, while datasets like UrbanSound8K focus on urban acoustic scenes, the aSwine dataset spans both human-related events (e.g., *Feed Provision*, *Speech*, or *Cleaning*) and pig-related events (e.g., *Stress/Disputes*, *Coughing/Sneezing*), reflecting a broader variety of real-world livestock scenarios.

As such, aSwine fills a critical gap in precision livestock audio mining, bringing attention to tasks like behavior monitoring and farm auditing.

4 Experimental setup

We performed the experiments on a single machine with the following configuration: RYZEN(R) 9@4.6GHz CPU, 32GB RAM, NVIDIA RTX 3090 24GB RAM GPU, on a Linux Pop!OS 22.04 LTS. We ran the audio features extractors using *librosa* [36], and ML/data utilities with *scikit-learn* [37] and *Pandas* [38]. Additionally, we implemented the deep learning models and training using *PyTorch* [39] and *PyTorch Lightning* [40], respectively. We illustrate the outline of our AED experimental framework in Fig. 4, which we detail in the following subsections.

Fig. 4 Overview of our AED experimental framework, from data collection to event detection. As this is a multilabel task, multiple events could be detected in a single audio sample



4.1 Data splitting and preprocessing

We divide aSwine the dataset into 80% for training and 20% for testing/validation. Then, we divide each audio file into 1-second segments and resample them to 16KHz. As such, each annotation in the dataset refers to 5 audio samples, thus totaling 43 817 and 10 979 samples for training and testing, respectively. We extract the log-mel spectrogram of each 1-second segment, with 25ms windows, 10ms hops (i.e., with an overlap of 15ms between each frame), 64 mel-spaced frequency bins, and a log offset of $1e - 2$, framed into a shape of 96×64 . Note that, in this process, we discard the final 15ms of information for each 1-second segment.

These 96×64 patches form the input to all models. Before employing deep learning methods, we Z-normalize the train and test/validation datasets according to the training dataset's mean and standard deviation.

4.2 Architectures and hyperparameter tuning

In the AED model training phase, we evaluated 6 different architectures: Fully Connected (as baseline), AlexNet [27], VGG [28], Inception V3 [29], ResNet-50 [30], and AST [26]. Our methodology is similar to the approach of Hershey et al. [21]. In the following subsections, we summarize the design choices for each architecture and outline how hyper-

parameters (e.g., depth, number of units, learning rates) were selected.

Fully Connected (FC)

For the FC architecture, we performed a grid search to find the optimal depth N and the number of units M per layer. Specifically, we tested $N \in \{2, 3, 4, 5, 6\}$ and $M \in \{512, 1024, 2048, 4096, 8192\}$. We employed ReLU activations [41], dropout at 0.5, and batch normalization across all layers. The final layer output dimension was 7, reflecting our seven event classes (multilabel).

AlexNet

We modified the original AlexNet [27] architecture's first convolutional layer to comply with our 96×64 inputs while also changing the stride and padding to 2×1 and 2×2 , respectively, to maintain a similar number of activations. Additionally, we used batch normalization after each convolutional layer and replaced the final fully connected layer with a 7 unit layer. We employed dropout layers with a 0.5 factor before the first two dense layers. This modified version of the AlexNet architecture has approximately 57M trainable parameters against the original architecture's 60M.

VGG

We applied VGG's configuration E [28] with the following adjustments: we modified the first convolutional layer to comply with our inputs, with batch normalization after each convolution in the architecture, and the final fully connected layer to comply with our 7 classes. The modified architecture has 139M trainable parameters against 144M in the original architecture.

Inception V3

We modified the Inception V3 [29] architecture by removing the first four layers and the auxiliary classifier network. To maintain the number of activations, we changed the average pool size to 10×6 . Additionally, we used batch normalization after each convolution layer. The resulting architecture decreases from the original's 26M to 22.6M trainable parameters.

ResNet-50

We employed ResNet-50 [30] and modified the architecture to maintain the number of activations by removing the stride of 2 from the first 7×7 convolution layer. We applied batch normalization after each convolutional layer. Additionally, we changed the average pool size to 6×4 . The modified network has 23.8M trainable parameters against the original's 26M.

Audio Spectrogram Transformer (AST)

We used the original AST [26] implementation¹ in its *base384* variant with a stride of 10 for both time and frequency dimensions. We did not modify the architecture itself to reach higher performance metrics; instead, we utilized pre-trained weights from ImageNet and AudioSet, as in the original AST paper. This allowed us to compare transformer-based performance with the same data and evaluation protocols used for the other architectures.

4.3 Training and evaluation

For the CNN-based architectures (AlexNet, VGG, Inception V3, ResNet-50), as well as the FC baseline (selected via grid search), we adopted AdamW [42] with an initial learning rate of 1×10^{-4} and weight decay of 1×10^{-2} . We empirically determined these values to balance training stability and generalization. By contrast, for AST, we used a smaller initial learning rate (1×10^{-5}) and weight decay (5×10^{-7}) to reflect its pre-trained starting point. Each model was trained with a mini-batch size of 64.

We relied on PyTorch's *BCEWithLogitsLoss*, which combines a Sigmoid activation with binary cross-entropy for multilabel classification. A dropout ratio of 0.5 was consistently applied to hidden layers, while early stopping interrupted training if validation loss ceased improving for 16 epochs. We also incorporated a learning rate scheduler that multiplied the learning rate by 0.1 whenever validation loss plateaued for more than 8 epochs. We stored checkpoint weights corresponding to the best validation ROC AUC observed during training.

Evaluation Metrics

We assessed each model using ROC AUC and Mean Average Precision (mAP). The area under the ROC curve (AUC) serves as a threshold-independent gauge of separability between positive and negative classes [43], with 1.0 indicating perfect discrimination and 0.5 approximating random guesses. Mean Average Precision (mAP), on the other hand, captures how effectively each model recovers true positives while avoiding false positives [44]. We used a class-specific prediction threshold of 0.5. Subsequently, we computed a *weighted* average of both AUC and mAP using the support of each of our seven event labels, thus accounting for the imbalance in class frequencies. Moreover, to gain finer-grained insight, we also report per-class AUC and mAP values in Section 5, demonstrating how specific events (e.g., *Others*

¹ Available at <https://github.com/YuanGongND/ast>

(H)) pose higher detection challenges than more frequent ones.

By combining grid-searched hyperparameter configurations (for the FC baseline) with empirically chosen parameters (for CNN and Transformer models), we ensured that each architecture was optimized reasonably for our domain. The following section presents the outcomes of these experiments and discusses the implications of each model's performance on AED tasks in livestock environments.

5 Results and discussion

This section presents a thorough discussion of our experimental findings, highlighting both quantitative results and qualitative insights. We address key questions concerning model performance and the roles of dataset characteristics, such as class imbalance, on the ability of different architectures to detect swine farm audio events accurately.

5.1 Overall performance comparison

We showcase the results of the best configurations for each method considering both AUC and mAP metrics in Tables 2 and 3, respectively. Our baseline results show that the best FC configuration was with 8 192 units per layer, reaching AUC and mAP scores of 0.817 (with 6 layers, and after 2.8h) and 0.735 (with 5 layers, and after 2.4h), respectively. Note that, to achieve the reported results, the baseline models required longer training than all other assessed architectures. Almost all the other evaluated architectures bested the baseline in both metrics, with the sole exception being VGG in terms of mAP.

We attained the highest AUC score of 0.850 using an AST model after 3 424 training steps, reaching a mAP score of 0.752 (Table 2). On the other hand, the highest mAP score was obtained by AlexNet (0.826), trained for 6 849 steps (Table 3), indicating that the models with the highest AUC scores may not necessarily achieve the highest mAP scores,

Table 2 Comparison of performance metrics for each evaluated architecture on the validation dataset with the best achieved AUC values. Bold values represent the best achieved metrics for each column

Architecture	Steps	Time	AUC	mAP
FC (6, 8 192)	21 234	2.8h	0.817	0.679
AlexNet	2 739	31.8min	0.841	0.733
VGG	2 739	38.6min	0.840	0.609
Inception V3	5 479	1.5h	0.848	0.755
ResNet-50	5 479	1.5h	0.822	0.687
AST	3 424	37.0min	0.850	0.752

Table 3 Comparison of performance metrics for each evaluated architecture on the validation dataset with the best achieved mAP values. Bold values represent the best achieved metrics for each column

Architecture	Steps	Time	AUC	mAP
FC (5, 8 192)	19 179	2.4h	0.811	0.735
AlexNet	6 849	1.8h	0.829	0.826
VGG	5 479	1.5h	0.838	0.678
Inception V3	4 794	1.3h	0.844	0.816
ResNet-50	6 164	1.7h	0.815	0.773
AST	2 739	27.7min	0.847	0.757

and vice versa. As we will discuss, these discrepancies likely reflect each model's sensitivity to class imbalance and noisy labels.

Additionally, we note that Inception V3 emerged as a strong competitor (AUC = 0.848, mAP = 0.755 in Table 2) despite having fewer parameters than VGG. This outcome aligns with existing studies suggesting that inception-based architectures can often learn complex audio patterns while being more computationally efficient [21]. Taken together, our findings reveal that architectures inspired by computer vision (AlexNet and Inception V3) and attention-based models (AST) can both excel in AED tasks. Yet, as we further explore in Section 5.2, each approach exhibits different strengths depending on event frequency and label noise.

5.2 Analysis of class-wise performance

For a deeper understanding of performance, we evaluate the AUC and mAP scores *per class* since our dataset exhibits significant imbalance (Fig. 2). Per-class metrics highlight the challenges of detecting particular events. Figures 5 and 6 show the best achieved AUC scores per class and trained

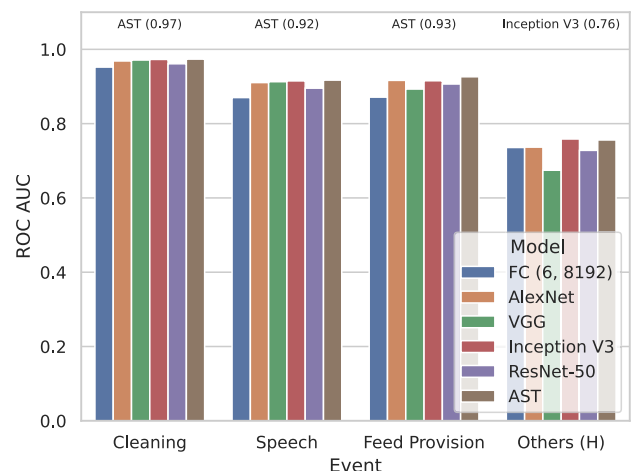


Fig. 5 AUC scores per event class and model for human events, emphasizing the best model per class

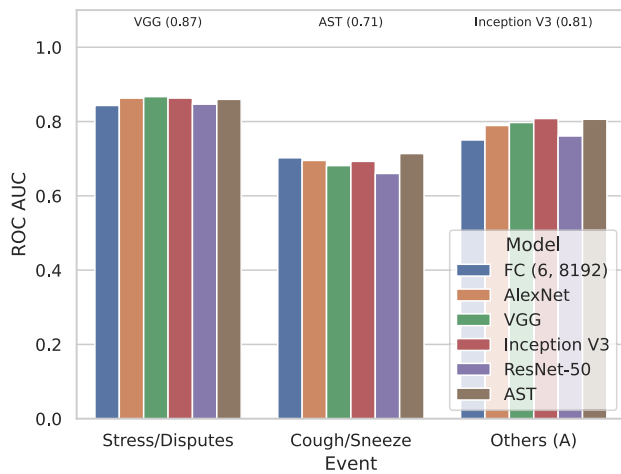


Fig. 6 AUC scores per event class and model for animal events, emphasizing the best model per class

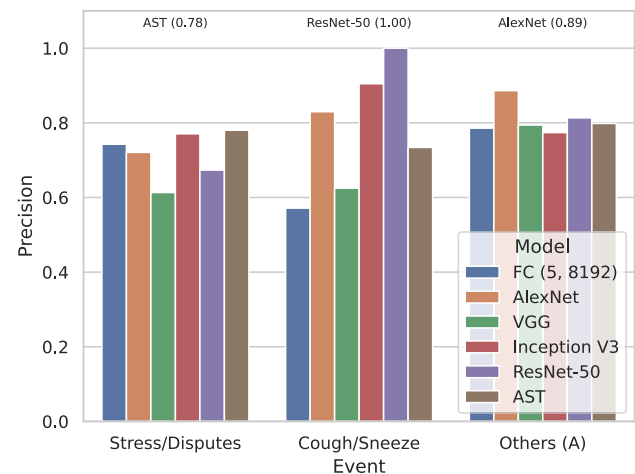


Fig. 8 mAP scores per event class and model for animal events, emphasizing the best model per class

model for human and animal events, respectively, while Figs. 7 and 8 show the best attained mAP scores.

AUC vs. mAP Under Class Imbalance

Although both AUC and mAP are used to evaluate our models in a class-weighted manner, they emphasize different aspects of predictive performance. AUC is threshold-independent and reflects how well a model ranks positive examples above negatives across all possible operating points [43], thereby tending to be more robust to imbalanced data when considered in aggregate. Meanwhile, mAP focuses on the precision-recall characteristics *at a chosen threshold* for each class and is, therefore, more sensitive to false positives in rare classes [44]. Although weighting by class support does mitigate some effects of imbalance, mAP can still exhibit high volatility for less frequent labels (e.g., *Feed Provision*) since even a few misclassifications can sharply reduce precision.

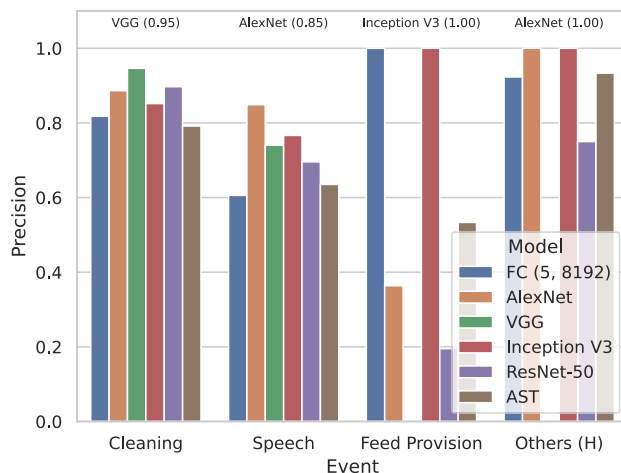


Fig. 7 mAP scores per event class and model for human events, emphasizing the best model per class

By jointly analyzing weighted AUC and mAP, along with per-class metrics, we gain a more comprehensive picture of model behavior: a model can display an excellent ability to separate positive from negative instances overall (high AUC) yet may struggle to maintain strong precision in classes with limited data (lower mAP).

Regarding human events, the *Others (H)* class was consistently more difficult to identify (maximum AUC 0.76 with Inception V3), likely due to its semantic diversity (e.g., random noises, non-vocal human sounds). This underscores the importance of more granular labeling or additional training examples for rare or heterogeneous classes. Conversely, more frequent events, such as *Cleaning* or *Speech*, tended to have higher mAP scores due to an increased number of positive training samples, which is consistent with well-known patterns in multilabel classification [45].

Interestingly, for animal events, our results (Fig. 6) show that detecting *Stress/Disputes* yielded relatively higher AUC values (max: 0.87 with VGG) than *Others (A)* or *Cough/Sneeze*. This may be attributed to the distinct acoustic characteristics of fighting or agitation sounds, which have higher energy in specific frequency bands. However, these classes also exhibited relatively lower mAP scores in some runs, revealing that the models occasionally produced more false positives when trying to capture such high-intensity events.

As shown in Figs. 7 and 8, mAP values were substantially affected by class imbalance, especially for *Feed Provision* and *Others (H)*. Because mAP is computed solely on positive examples and is sensitive to false positives, rarer events can yield volatile precision values. In contrast, AUC incorporates both positive and negative examples and can, therefore, mask certain precision issues [43]. This highlights the necessity for

combining multiple metrics to gain a comprehensive view of performance in weakly-labeled and imbalanced settings.

5.3 Influence of data characteristics and model design

The discussed results show that class imbalance and noisy, open-environment recordings pose challenges for the models. For instance, while noise-reduction techniques could theoretically alleviate some background sounds, heavy filtering can risk losing critical signal characteristics. This trade-off is particularly pronounced for subtle classes (e.g., *Cough/Sneeze*), where high-frequency components can be easily suppressed. Moreover, overlapping events transform AED into a multilabel learning problem, adding substantial complexity beyond standard single-label classification tasks.

The AST model's ability to capture long-range temporal dependencies may partly explain why it excelled in AUC, but it required pre-training on larger datasets. This confirms the consensus in the literature that attention-based networks often demand more extensive data to outperform convolutional counterparts [26]. Nonetheless, AlexNet and Inception V3, trained from scratch, achieved competitive mAP scores, suggesting that simpler architectures can remain effective when augmented by domain knowledge (e.g., spectrogram design, specialized data splits).

From a practical standpoint, these findings imply that for real-world deployment in livestock farms, practitioners may choose AlexNet or Inception V3 if computation and annotation resources are limited, given their strong performance *without* large-scale pre-training. However, for users who can gather larger annotated corpora or leverage transfer learning from massive audio datasets (e.g., AudioSet), attention-based architectures such as AST might yield more robust event detection under varying conditions.

Finally, we note that the relatively small size and class imbalance in our dataset suggest caution when generalizing these results. Larger-scale studies could further boost event detection accuracies, potentially integrating data augmentation or semi-supervised strategies. Additionally, future experiments could explore ensemble methods that combine CNN and Transformer-based features, aiming to capture both local (convolutional) and global (attention-based) properties of audio signals.

6 Conclusions

In conclusion, our study represents a significant step forward in exploring audio data for potential auditing methods in ani-

mal farms. The underutilized potential of sound data and automation in PLF industrial settings has been underscored, with various promising applications and advancements yet to be fully realized. Overall, our results indicate:

- AST attained the highest AUC (0.850), while AlexNet reached the highest mAP (0.826). Despite AST's reliance on larger-scale pre-training for optimal performance, its attention mechanism enabled superior global context modeling, leading to the best AUC. AlexNet, by contrast, showed strong feature extraction and generalization even without extensive pre-training, ultimately achieving the highest mAP in our experiments.
- Data imbalance and class-specific characteristics significantly affect performance. Classes with few samples (e.g., *Feed Provision*) or broad semantic diversity (e.g., *Others (H)*) posed greater challenges, disproportionately lowering mAP. This underscores the necessity for careful data collection or specialized strategies (e.g., class weighting) to mitigate performance drops.
- CNN architectures can remain competitive on small datasets without pre-training. Even in noisy, open-environment recordings, modified CNN-based models such as AlexNet and Inception V3 demonstrated robust performance, indicating that domain-specific architectures and parameter tuning can compensate for the lack of large-scale pre-training data. Nevertheless, recent advances in transformer models, such as self-supervision strategies, may reduce the reliance on pre-training on large datasets [46], thus presenting a potential direction for future investigation.
- Model ensembles offer the potential for further gains. Preliminary findings suggest that ensembling CNN and attention-based methods may combine complementary strengths – local pattern extraction and global context modeling – thus paving the way for improved event detection across challenging classes.

Our work paves the way for further studies in several directions. First, increasing the size and diversity of labeled data can benefit attention-based models. However, given the potential scarcity of large-scale datasets in livestock contexts, further optimization of architectures like AST may focus on more lightweight variants or techniques such as partial fine-tuning (e.g., freezing lower-layer parameters) to reduce data requirements. At the same time, advanced learning strategies (e.g., active learning, semi-supervised, and self-supervised approaches) may reduce labeling costs and improve coverage of rare classes. Advanced data augmentation techniques, such as time-frequency masking, pitch shifting, and synthetic

overlapping acoustic events, could help models become more robust to real-world noise. Ensemble methods that merge CNN and Transformer-based architectures may combine local pattern extraction and global context modeling strengths, yielding further performance gains. From a practical standpoint, evaluating knowledge distillation strategies can lead to less resource-intensive models, which is crucial for meeting the real-time and hardware constraints often encountered in livestock environments. Additionally, exploring small-scale attention networks or pruning techniques may enable attention-based architectures to retain strong performance under limited data scenarios.

Finally, exploring multimodal approaches that integrate audio with other sensing modalities – like video or environmental parameters – could foster a more comprehensive perspective in precision livestock farming. Importantly, these methodologies may also extend to other domains managing complex, real-world data. For instance, the same strategies we employ to handle noisy inputs and weak labels in livestock settings could be adapted for medical imaging tasks like voxel-wise body composition analysis from limited CT slices [47], thus underlining the broader applicability of our work. Overall, these directions underscore our belief that audio-based monitoring holds promise not only for improving farming practices, optimizing resource utilization, and advancing animal welfare but also for inspiring cross-domain solutions in similarly challenging data scenarios.

The findings in our study validate the feasibility of leveraging continuous audio monitoring to detect critical events such as coughing (potentially indicating health issues) or stress behaviors (e.g., vocalizations during fights). Real-time alerts could be integrated into farm management systems, allowing faster intervention, reduced animal suffering, and more efficient resource allocation. Moreover, human-related events (e.g., *Cleaning*, *Feed Provision*, *Speech*) provide an additional layer of auditing, ensuring compliance with best practices and regulatory standards. By adopting these audio-based tools, veterinarians and farm managers can better assess handling procedures, promptly address anomalies, and ultimately enhance animal welfare. As we conclude, we advocate for increased attention from researchers in veterinary and livestock fields towards incorporating audio data in their future studies, fostering further advancements in animal monitoring and welfare.

Acknowledgements This research was supported by FAPESP (São Paulo State Research Foundation) (grants 2016/19514-2, 2016/17078-0, and 2020/07200-9), CAPES (Brazilian Coordination for Improvement of Higher Level Personnel) (grants PROEX-9778985/M and 88887.893807/2023-00), and CNPq (Brazilian National Council for Supporting Research) (grant 4088672021-3).

Availability of data and materials The Annotated Swine (aSwine) dataset's audio files and annotations used in this study are publicly available at <https://github.com/andremsouza/aswine>.

Code Availability The source code for this research is publicly available at <https://github.com/andremsouza/vision-aed-swine-barn-weak-labels>.

Declarations

Competing Interests The authors declare no conflict of interest.

Ethics approval Research on animals was conducted according to the institutional committee on animal use (protocol nº 3219080822).

References

1. Rojas-Downing MM, Nejadhashemi AP, Harrigan T, Woznicki SA (2017) Climate change and livestock: impacts, adaptation, and mitigation. *Clim Risk Manag* 16:145–163. <https://doi.org/10.1016/j.crm.2017.02.001>
2. Schillings J, Bennett R, Rose DC (2021) Exploring the potential of precision livestock farming technologies to help address farm animal welfare. *Front Anim Sci* 2. <https://doi.org/10.3389/fanim.2021.639678>
3. Chung Y, Oh S, Lee J, Park D, Chang H-H, Kim S (2013) Automatic detection and recognition of pig wasting diseases using sound data in audio surveillance systems. *Sensors* 13(10):12929–12942. <https://doi.org/10.3390/s131012929>
4. Benjamin M, Yik S (2019) Precision livestock farming in swine welfare: a review for swine practitioners. *Animals* 9(4). <https://doi.org/10.3390/ani9040133>
5. Norton T, Chen C, Larsen MLV, Berckmans D (2019) Review: precision livestock farming: building 'digital representations' to bring the animals closer to the farmer. *Animal* 13(12):3009–3017. <https://doi.org/10.1017/S175173111900199X>
6. Neethirajan S (2020) The role of sensors, big data and machine learning in modern animal farming. *Sens Bio-Sens Res* 29:100367. <https://doi.org/10.1016/j.sbsr.2020.100367>
7. Yin Y, Tu D, Shen W, Bao J (2021) Recognition of sick pig cough sounds based on convolutional neural network in field situations. *Inf Process Agric* 8(3):369–379. <https://doi.org/10.1016/j.inpa.2020.11.001>
8. Atrey PK, Maddage NC, Kankanhalli MS (2006) Audio based event detection for multimedia surveillance. In: 2006 IEEE international conference on acoustics speech and signal processing proceedings, vol 5. <https://doi.org/10.1109/ICASSP.2006.1661400>
9. Portelo J, Bugalho M, Trancoso I, Neto J, Abad A, Serralheiro A (2009) Non-speech audio event detection. In: 2009 IEEE international conference on acoustics, speech and signal processing, pp 1973–1976. <https://doi.org/10.1109/ICASSP.2009.4959998>
10. Lu L, Ge F, Zhao Q, Yan Y (2010) A SVM-based audio event detection system. In: 2010 international conference on electrical and control engineering, pp 292–295. <https://doi.org/10.1109/ICECE.2010.78>
11. Kumar A, Raj B (2016) Audio event detection using weakly labeled data. In: Proceedings of the 24th ACM international conference on multimedia. MM '16, pp. 1038–1047. Association for Computing Machinery, New York, NY, USA. <https://doi.org/10.1145/2964284.2964310>
12. Ahmad N, Strand R, Sparresäter B, Tarai S, Lundström E, Bergström G, Ahlström H, Kullberg J (2023) Automatic segmentation of large-scale CT image datasets for detailed body composition analysis. *BMC Bioinformatics* 24(1):1–21. <https://doi.org/10.1186/s12859-023-05462-2>
13. Ahmad N, Asghar S, Gillani SA (2022) Transfer learning-assisted multi-resolution breast cancer histopathological images classi-

- fication. *Vis Comput* 38(8):2751–2770. <https://doi.org/10.1007/s00371-021-02153-y>
14. Malik M, Malik MK, Mehmood K, Makhdoom I (2020) Automatic speech recognition: a survey. *Multimed Tools Appl* 80(6):9411–9457. <https://doi.org/10.1007/s11042-020-10073-7>
15. Mesaros A, Heittola T, Virtanen T, Plumbley MD (2021) Sound event detection: a tutorial. *IEEE Signal Process Mag* 38(5):67–83. <https://doi.org/10.1109/MSP.2021.3090678>
16. Yan C, Shunyuan L, Yulong Q (2020) Classification of pig sounds based on deep neural network. *Trans Chin Soc Agric Eng (Transactions of the CSAE)* 36(9):195–204. <https://doi.org/10.11975/j.issn.1002-6819.2020.09.022>
17. Alves AAC, Andrietta LT, Lopes RZ, Bussiman FO, Silva FFe, Carnevalheiro R, Brito LF, Balieiro JcDc, Albuquerque LG, Ventura RV (2021) Integrating audio signal processing and deep learning algorithms for gait pattern classification in brazilian gaited horses. *Front Anim Sci* 2. <https://doi.org/10.3389/fanim.2021.681557>
18. Wang Y, Li S, Zhang H, Liu T (2022) A lightweight CNN-based model for early warning in sow oestrus sound monitoring. *Eco Inform* 72:101863. <https://doi.org/10.1016/j.ecoinf.2022.101863>
19. Laguna EB, Mun H-S, Ampode KMB, Chem V, Kim Y-H, Yang C-J (2023) Artificial intelligence for automatic monitoring of respiratory health conditions in smart swine farming. *Animals* 13(11):1860. <https://doi.org/10.3390/ani13111860>
20. Ding B, Zhang T, Wang C, Liu G, Liang J, Hu R, Wu Y, Guo D (2024) Acoustic scene classification: a comprehensive survey. *Expert Syst Appl* 238:121902. <https://doi.org/10.1016/j.eswa.2023.121902>
21. Hershey S, Chaudhuri S, Ellis DPW, Gemmeke JF, Jansen A, Moore RC, Plakal M, Platt D, Saurous RA, Seybold B, Slaney M, Weiss RJ, Wilson K (2017) CNN architectures for large-scale audio classification. In: 2017 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), pp 131–135. <https://doi.org/10.1109/ICASSP.2017.7952132>
22. Su T-W, Liu J-Y, Yang Y-H (2017) Weakly-supervised audio event detection using event-specific gaussian filters and fully convolutional networks. In: 2017 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), pp 791–795. <https://doi.org/10.1109/ICASSP.2017.7952264>
23. Mac Aodha O, Gibb R, Barlow KE, Browning E, Firman M, Freeman R, Harder B, Kinsey L, Mead GR, Newson SE, Pandourski I, Parsons S, Russ J, Szodoray-Paradi A, Szodoray-Paradi F, Tilova E, Girolami M, Brostow G, Jones KE (2018) Bat detective-deep learning tools for bat acoustic signal detection. *PLoS Comput Biol* 14(3):1–19. <https://doi.org/10.1371/journal.pcbi.1005995>
24. Morfi V, Stowell D (2018) Deep learning for audio event detection and tagging on low-resource datasets. *Appl Sci* 8(8). <https://doi.org/10.3390/app8081397>
25. Kong Q, Xu Y, Wang W, Plumbley MD (2020) Sound event detection of weakly labelled data with CNN-Transformer and automatic threshold optimization. *IEEE/ACM Trans Audio Speech Lang Process* 28:2450–2460. <https://doi.org/10.1109/TASLP.2020.3014737>
26. Gong Y, Chung Y-A, Glass J (2021) AST: audio spectrogram transformer. In: *Proc Interspeech* 2021, pp 571–575. <https://doi.org/10.21437/Interspeech.2021-698>
27. Krizhevsky A, Sutskever I, Hinton GE (2017) ImageNet classification with deep convolutional neural networks. *Commun ACM* 60(6):84–90. <https://doi.org/10.1145/3065386>
28. Simonyan K, Zisserman A (2015) Very deep convolutional networks for large-scale image recognition. <https://doi.org/10.48550/arXiv.1409.1556>
29. Szegedy C, Vanhoucke V, Ioffe S, Shlens J, Wojna Z (2016) Rethinking the inception architecture for computer vision. In: 2016 IEEE Conference on Computer Vision and Pattern Recognition (CVPR), pp 2818–2826. <https://doi.org/10.1109/CVPR.2016.308>
30. He K, Zhang X, Ren S, Sun J (2016) Deep residual learning for image recognition. In: 2016 IEEE Conference on Computer Vision and Pattern Recognition (CVPR), pp 770–778. <https://doi.org/10.1109/CVPR.2016.90>
31. Gemmeke JF, Ellis DPW, Freedman D, Jansen A, Lawrence W, Moore RC, Plakal M, Ritter M (2017) Audio set: an ontology and human-labeled dataset for audio events. In: 2017 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), pp 776–780. <https://doi.org/10.1109/ICASSP.2017.7952261>
32. Zhang T, Zhang K, Wu J (2018) Temporal transformer networks for acoustic scene classification. In: *Proc Interspeech* 2018, pp 1349–1353. <https://doi.org/10.21437/Interspeech.2018-1152>
33. Ren Z, Kong Q, Han J, Plumbley MD, Schuller BW (2021) CAA-Net: conditional atrous CNNs with attention for explainable device-robust acoustic scene classification. *IEEE Trans Multimedia* 23:4131–4142. <https://doi.org/10.1109/TMM.2020.3037534>
34. Dosovitskiy A, Beyer L, Kolesnikov A, Weissenborn D, Zhai X, Unterthiner T, Dehghani M, Minderer M, Heigold G, Gelly S, Uszkoreit J, Houshy N (2021) An image is worth 16x16 words: transformers for image recognition at scale. <https://doi.org/10.48550/arXiv.2010.11929>
35. Salamon J, Jacoby C, Bello JP (2014) A dataset and taxonomy for urban sound research. In: *Proceedings of the 22nd ACM international conference on multimedia*. MM '14. Association for Computing Machinery, New York, NY, USA, pp 1041–1044. <https://doi.org/10.1145/2647868.2655045>
36. McFee B, Raffel C, Liang D, Ellis DP, McVicar M, Battenberg E, Nieto O (2015) librosa: audio and music signal analysis in python. In: Huff, Bergstra (eds) *Proceedings of the 14th python in science conference*, pp 18–24. <https://doi.org/10.25080/Majora-7b98e3ed-003>
37. Pedregosa F, Varoquaux G, Gramfort A, Michel V, Thirion B, Grisel O, Blondel M, Prettenhofer P, Weiss R, Dubourg V, Vanderplas J, Passos A, Cournapeau D, Brucher M, Perrot M, Duchesnay E (2011) Scikit-learn: Machine learning in python. *J Mach Learn Res* 12(85):2825–2830. <https://doi.org/10.5555/1953048.2078195>
38. McKinney W (2010) Data structures for statistical computing in python. In: Walt, Millman (eds) *Proceedings of the 9th python in science conference*, pp 56–61. <https://doi.org/10.25080/Majora-92bf1922-00a>
39. Paszke A, Gross S, Massa F, Lerer A, Bradbury J, Chanan G, Killeen T, Lin Z, Gimelshein N, Antiga L, Desmaison A, Köpf A, Yang E, DeVito Z, Raison M, Tejani A, Chilamkurthy S, Steiner B, Fang L, Bai J, Chintala S (2019) Pytorch: an imperative style, high-performance deep learning library. In: Wallach H, Larochelle H, Beygelzimer A, Alché-Buc F, Fox E, Garnett R (eds) *Advances in neural information processing systems*, vol 32. Curran Associates Inc., Red Hook, NY, USA. <https://doi.org/10.5555/3454287.3455008>
40. Falcon W (2019) The PyTorch lightning team: PyTorch lightning, 1.4. <https://doi.org/10.5281/zenodo.3828935>. <https://github.com/Lightning-AI/lightning>
41. Nair V, Hinton GE (2010) Rectified linear units improve restricted boltzmann machines. In: *Proceedings of the 27th international conference on international conference on machine learning*. ICML'10. Omnipress, Madison, WI, USA, pp 807–814. <https://doi.org/10.5555/3104322.3104425>
42. Loshchilov I, Hutter F (2019) Decoupled weight decay regularization. <https://doi.org/10.48550/arXiv.1711.05101>
43. Huang J, Ling CX (2005) Using auc and accuracy in evaluating learning algorithms. *IEEE Trans Knowl Data Eng* 17(3):299–310. <https://doi.org/10.1109/TKDE.2005.50>

44. Hossin M, Sulaiman MN (2015) A review on evaluation metrics for data classification evaluations. *Int J Data Mining Knowl Manag Process* 5(2):01–11. <https://doi.org/10.5121/ijdkp.2015.5201>
45. Zhu Y, Kwok JT, Zhou Z-H (2018) Multi-Label Learning with Global and Local Label Correlation. *IEEE Trans Knowl Data Eng* 30(6):1081–1094. <https://doi.org/10.1109/TKDE.2017.2785795>
46. Gong Y, Lai C-I, Chung Y-A, Glass J (2022) SSAST: self-supervised audio spectrogram transformer. *Proc AAAI Conf Artif Intell* 36(10):10699–10709. <https://doi.org/10.1609/aaai.v36i10.21315>
47. Ahmad N, Dahlberg H, Jönsson H, Tarai S, Guggilla RK, Strand R, Lundström E, Bergström G, Ahlström H, Kullberg J (2024) Voxel-wise body composition analysis using image registration of a three-slice CT imaging protocol: methodology and proof-of-

concept studies. *Biomed Eng Online* 23(1):1–16. <https://doi.org/10.1186/s12938-024-01235-x>

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